Curv AI (Curv Health / Curv Labs Inc.) Tech Stack Deep Dive (April 2026)

Curv's technology is a **hybrid computer vision + machine learning + large language model (LLM) system** centered on smartphone video analysis. It powers their "AI Sport Scientist" — marketed as "ChatGPT for athletes, but it can *see* the athlete."

The core promise: Turn any smartphone video into objective biomechanical data (explosiveness, flexibility, movement efficiency, asymmetry, injury risk proxies, cheer-specific skills) → feed that structured data into an adaptive AI coach that generates personalized training plans, answers questions 24/7, and supports recruiting/leaderboards.

No official "tech stack" page or engineering blog exists (typical for competitive advantage in CV/ML). All details below are synthesized from:

- Official site, app store descriptions, PRs (2025), Betakit (2019), Youth Sports Business Report (2025)
- Executive quotes (Shea Balish)
- Team descriptions ("full-stack engineers, computer vision developers, machine-learning experts, elite biomechanists")
- Co-founder signals
- Industry patterns for similar 2017–2026 startups

High-Level Architecture

Input → Processing → Intelligence → Output

1. **Smartphone Video Capture** (user records jumps, skills, movements)
2. **Computer Vision Pipeline** (on-device + cloud) → extracts structured biomechanical features
3. **ML Inference Layer** → scores, risk flags, form analysis (cheer-specific + general athleticism)
4. **LLM + RAG Reasoning Layer** ("AI Sport Scientist") → conversational coach, adaptive plans, holistic advice (training + nutrition + recovery + mindset)

5. **Mobile App Layer** → assessments, leaderboards, profiles, Curv+ human expert booking
6. **Backend SaaS/API** (historical) + real-time features (leaderboards, updates)

Key Differentiator (per PRs): CV provides "ground truth" objective data that pure text LLMs lack. The system "improves with every interaction."

Development Timeline:

- 2017–2019: Foundational 2D video CV + ML for biomechanical feature extraction (rehab/wellness focus). "Machine learning tools... extract biomechanical features of human motion and shape from a two-dimensional video."
- 2022: Expanded remote monitoring (rep counting, at-home programs) + digital clinic.
- 2025: LLM fusion + "AI Sport Scientist" launch + dedicated cheer app. 6+ years of proprietary model refinement (NFL-funded early work).

Detailed Stack Components

1. Mobile Frontend / App Layer

- **Primary Framework**: **React Native** (very strong evidence — co-founder Nic Tancredi's professional profile explicitly lists React, React Native, ReactJS + native Swift/Xcode experience at Curv Labs Inc.).
- Native modules for camera/video (AVFoundation iOS, CameraX Android) and performance-critical CV.
- **iOS + Android** apps (Curv Labs Inc. developer). iOS app size ~154.5 MB (suggests bundled models, assets, or offline capabilities).
- Features implemented: Video recording/analysis, computer vision assessments (added in updates), progress tracking, leaderboards, recruiting profiles, highlight reel submissions, cheer-specific dashboard.
- Real-time elements: Leaderboards and updates (likely WebSockets or Firebase Realtime).

2. Computer Vision Core (The Foundation)

- **Input**: Standard smartphone video (no special hardware, any practical angle/lighting).
- **Capabilities** (confirmed):
  - Pose/landmark detection and tracking over time
  - Biomechanical metrics: explosiveness (power/velocity), flexibility (range of motion), movement efficiency, body asymmetry, injury risk indicators (e.g., compensatory patterns)
  - Cheer-specific: Jump analysis, strength/flexibility holds, skill form scoring
  - Additional: Rep counting, form feedback, highlight generation
- **Technologies (High-Probability Stack)**:
  - **Google MediaPipe** (Pose, Holistic, or custom solutions) — On-device, real-time, battery-efficient landmark detection. The de-facto standard for exactly this mobile body-tracking use case in 2025–2026.
  - **OpenCV** — Video preprocessing, filtering, custom feature extraction, temporal analysis.
  - **Custom ML Models**: PyTorch (training) or **TensorFlow Lite** / Core ML (on-device inference). Fine-tuned on proprietary athlete datasets + public biomechanics data. “Elite biomechanists” translate math into sport-specific metrics.
- **Processing Model**: Hybrid on-device (lightweight pose estimation for low latency/privacy) + cloud (heavier analysis, model updates, complex cheer metrics). Explains occasional loading complaints in app reviews.
- **Proprietary Edge**: 6+ years of custom training data and domain-specific fine-tuning (cheer stunts, landings, synchronization proxies). Not just off-the-shelf MediaPipe.

3. Machine Learning & Generative AI Layer (“AI Sport Scientist”)

- **Traditional ML**: Regression/classification models on pose time-series for objective scoring and risk flagging.

- **LLM Integration** (2025+): “Like ChatGPT... with a major leap forward” — CV outputs are injected into the LLM context.
  - **Base Model**: Almost certainly **OpenAI GPT-4o / o-series** (or successors) via API. Possible multi-model routing.
  - **RAG Architecture**: Vector retrieval over:
    - Curated knowledge base (vetted protocols from “top coaches, sports scientists, former collegiate athletes”)
    - User history + CV features
    - Research/biomechanics literature
  - **Vector Database**: Likely **pgvector** (Postgres), Pinecone, or Weaviate.
  - Output: Conversational coaching, adaptive training plans (updates in seconds based on feedback/injury/goals), nutrition/recovery/mindset advice.
- **Data Fusion**: Apple HealthKit integration (sleep, heart rate) for holistic profiles.
- **Safety/Quality**: “AI programs vetted” — human oversight layer + domain rules.

4. Backend, Data & Infrastructure

- **API/Server Framework**: Python ecosystem most likely (**FastAPI** preferred for ML-heavy teams; possible Django/Flask). Node.js possible given React Native lean.
- **Cloud Provider**: **AWS** (SageMaker for ML training/hosting, S3 for secure video storage, Lambda/EC2, Cognito) or **GCP** (Vertex AI — excellent for CV/ML workloads). Multi-cloud possible.
- **Databases**:
  - **PostgreSQL** (core: users, training plans, assessment scores, recruiting profiles)
  - **Redis** (real-time leaderboards, caching, sessions)
  - Possibly Firebase/Firestore for mobile-first sync
- **Storage & Processing**: Encrypted cloud video handling; on-device options where feasible.

- **Auth & Compliance**: JWT/OAuth; strong privacy emphasis ("everything stays between you and the platform"). Health-data adjacent controls.
- **Real-time & Scalability**: WebSockets or equivalent for live leaderboards/events. Designed for global use ("any athlete, anywhere").

5. Supporting / DevOps Layer (Inferred Standard)

- CI/CD: GitHub Actions + Docker
- Monitoring: Datadog, New Relic, or Sentry
- Analytics: Amplitude, Mixpanel, or Firebase
- Experimentation: Jupyter + MLflow or similar (team has ML experts)

Evidence Summary & Confidence Levels

- **Confirmed**: Advanced computer vision on smartphone video; ML for biomechanical features from 2D footage; LLM + CV hybrid; Apple HealthKit; proprietary models refined over 6+ years; React Native signals (co-founder skills).
- **High Confidence Inferences**: MediaPipe + OpenCV + custom PyTorch/TF Lite; OpenAI GPT + RAG; AWS/GCP + Postgres/Redis; FastAPI backend.
- **Gaps (No Public Info)**: Exact model versions, dataset sizes, training pipelines, specific vector DB, or proprietary algorithms. No patents identified for their CV methods. No engineering blog or open-source repos.

Maturity & Limitations (from app reviews + PRs):

- Strengths: Scalable (phone-only), objective metrics, continuously improving, strong privacy posture, validated at events/showcases.
- Observed Issues: Loading times for assessments (video processing bottleneck); some users find AI responses generic (possible RAG quality or prompt tuning gaps); early-stage app (only 13 iOS ratings).

Strategic Implications for Your Cheer Software

Curv's stack is **modern, pragmatic, and effective** for broad accessibility — exactly what a venture-backed startup with NFL/VC backing would build (MediaPipe + custom CV + LLM

Venture-backed startup with VC/PE backing would build (modular pipe – custom CV + LLM RAG is a 2025–2026 winning pattern).

Where You Can Win with "Much Deeper" AI Coach:

- **Specialized CV Models**: Go beyond general pose estimation. Train dedicated models for cheer-specific dynamics (multi-athlete interactions in stunts/pyramids, aerial trajectories, synchronization scoring, base/flyer roles, mat safety).
- **Domain Depth**: Richer proprietary cheer dataset + rules engine (progressions, spotting, competition strategy, team chemistry) feeding the LLM.
- **Superior Reasoning**: More advanced RAG, agentic workflows, or fine-tuned models tailored to cheer coaching nuances (vs. general athleticism).
- **Integration & UX**: Tighter coupling with your full software suite (team management, routines, video libraries) vs. standalone app.
- **Performance Edge**: Heavier on-device processing or optimized pipeline to beat their loading complaints.
- **Data Moat**: If you have deeper cheer-specific labeled data or coach partnerships, your models can outperform on niche metrics.

Curv has strong execution on the *foundation* (objective measurement + accessible AI). Your advantage lies in **vertical depth** within cheer.

This is the deepest publicly derivable technical analysis possible. If you share more about *your* current stack or specific areas (e.g., "compare our CV approach" or "help me design a superior RAG system"), I can provide targeted recommendations or even architecture diagrams. Ready to help you outbuild them! 🚀